

MICROCANONICAL PREPARATION AND EXPONENTIAL-SCALE ENSEMBLE TRANSFER FOR DETERMINISTIC HARD SPHERES

QIAN QI

ABSTRACT. We derive the source-dependent microcanonical denominator directly from the positive interacting exact-particle-number hard-sphere integral. No independence of connected components and no probabilistic interpretation of signed polymer activities are used. Sequential physical-cell exposure and velocity integration by parts give one integrable characteristic-function majorant over all frequencies, a uniformly positive covariance, and a geometric mixed-shell local coefficient theorem. The proof order is strictly grand-canonical B2, then B1 conditioning, then the microcanonical B2 large-deviation theorem.

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CONTENTS

1. Round-twenty-one positive closure: fixed-size smoothing blocks and exact-number saddle extraction	2
1.1. Constraint space and the source-dependent saddle	2
1.2. Fixed-size full-rank smoothing blocks	3
1.3. A global characteristic majorant without $O(N)$ derivatives	4
1.4. Mixed lattice-continuous coefficient asymptotics	5
1.5. Exact-number extraction from the B2 pressure	5
Scope and verification status	6

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1. ROUND-TWENTY-ONE POSITIVE CLOSURE: FIXED-SIZE SMOOTHING
BLOCKS AND EXACT-NUMBER SADDLE EXTRACTION

This section replaces the former Round-Seventeen source. The scaling is the three-dimensional Boltzmann–Grad regime of B2: $\mu_\varepsilon = \varepsilon^{-2}$ and $N/\mu_\varepsilon \rightarrow \rho$ in a fixed compact subset of $(0, \infty)$. The finite-volume law is the positive exact- N hard-sphere phase-space integral. Connected activities are used only as analytic estimates for this positive integral and are never interpreted as independent random components.

1.1. **Constraint space and the source-dependent saddle.** Let

$$C(x, v) = (v, |v|^2/2, \chi_1(x), \dots, \chi_m(x)),$$

where the smooth cell functions χ_j have compactly supported gradients in pairwise separated interior boxes. Let $\mathcal{R}$ be the finite-dimensional space of affine relations among the components of C and put $E = \mathbb{R}^d/\mathcal{R}$. All covariances and Fourier variables below are on E ; redundant partition-of-unity coordinates are not assigned a positive variance.

For a bounded regular path/contact source H and multiplier $\lambda \in E^*$, define

$$Z_{\varepsilon, N}(H, \lambda) = \frac{1}{N!} \int_{\mathcal{D}_{\varepsilon, N} \times \mathbb{R}^{3N}} \exp \left\{ \sum_{i=1}^N \lambda \cdot C(x_i, v_i) + \mu_\varepsilon H(\mathbf{X}) \right\} d\mathbf{x} d\mathbf{v}.$$

The source class is the same analytic ball used in the B2 pressure theorem.

Theorem 1.1 (Uniform canonical pressure and saddle). *There is an open convex multiplier domain $\mathcal{O} \subset E^*$ such that*

$$q_N(H, \lambda) = \frac{1}{N} \log Z_{\varepsilon, N}(H, \lambda)$$

converges, with three multiplier derivatives and two source derivatives, locally uniformly to a strictly convex function $q(H, \lambda)$. On every compact set A in the interior of its mean image, the equation

$$D_\lambda q(H, \lambda) = a$$

has a unique solution $\lambda(H, a)$, analytic in H and a , and

$$cI_E \leq D_\lambda^2 q(H, \lambda(H, a)) \leq CI_E.$$

The same estimates hold for q_N and its saddle for all sufficiently small ε .

Proof. Insert an activity variable z in the B2 grand-canonical pressure and use its normally convergent connected series on a short time slice. The coefficient of z^N is exactly the positive integral above. Uniform convergence on a complex annulus gives the canonical logarithmic limit and permits the stated source and multiplier derivatives. At zero interaction, the log moment of C is strictly convex on the quotient E because no nonzero element of E^* is constant on the support. The B2 connected remainder has Hessian $O(\varepsilon)$

uniformly on compact multiplier sets; hence the lower and upper covariance bounds persist.

For an exponential family on a compact multiplier chart, the gradient of a strictly convex log partition is a diffeomorphism onto the interior of the convex support. Properness follows because approaching the chart boundary concentrates on an exposed face, while A stays a positive distance from all faces. The interacting gradient is a uniform small perturbation and has the same degree. Thus it is globally one-to-one and onto A , not merely locally invertible. The Banach analytic implicit-function theorem gives dependence on H and a . $\square$

1.2. Fixed-size full-rank smoothing blocks. Choose $b = d + 2$ pairwise separated position boxes and velocity balls so that the block map

$$\mathcal{C}_B : (x_1, v_1, \dots, x_b, v_b) \mapsto \sum_{j=1}^b C(x_j, v_j) \in E$$

has a d -dimensional minor bounded below on a compact subbox. Position variables are included in this minor whenever the Fourier direction is position dominated. The boxes are separated by a fixed distance, so no hard-core boundary occurs inside the block chart.

A block is called fresh if its particles have no prescribed collision with previously exposed particles during the short preparation layer and their initial data lie in one of the full-rank charts. Expose blocks successively, conditioning on the exterior configuration.

Lemma 1.2 (Uniform conditional block regularity). *There are an integer $s > d+5$, constants $c, C > 0$, and $\alpha > 0$ such that the following holds. Except on an event of probability at most e^{-cN} , at least αN disjoint fresh blocks can be selected. Conditional on the exterior and on all previously selected blocks, the next block has density*

$$p_B = p_B^{(0)} R_B$$

on a full-rank chart, where $p_B^{(0)}$ is a product density bounded above and below and

$$\sum_{|\gamma| \leq s} \|\partial_B^\gamma R_B\|_{L^1} \leq C.$$

The constants are uniform in N, ε, H , and the saddle.

Proof. Tile phase space by a fixed finite family of separated position–velocity boxes. Hard-core exclusion removes $O(N\varepsilon^3) = O(\varepsilon)$ of the conditional position volume because $N \asymp \varepsilon^{-2}$. The B2 connected expansion with one block of marked root variables gives, after conditioning on the exterior, a Radon–Nikodym factor whose derivatives in that fixed block are bounded by the normally convergent marked cluster series. The block size b and derivative order s are fixed; no derivative order grows with N . The dynamic source is differentiated only through clusters meeting the block, and the B2 source ball gives the same bound.

On each exposure, either a full-rank box is available with conditional probability at least $p_* > 0$, or the configuration lies in a union of forbidden-volume and singular-coordinate events of conditional probability $O(\varepsilon)$. A martingale Chernoff bound for these bounded conditional success indicators gives at least αN successes with probability $1 - e^{-cN}$. Taking the finite union over the block atlas proves the lemma. $\square$

Lemma 1.3 (One-block Fourier smoothing). *For every conditional block density in Lemma 1.2,*

$$\left| \int e^{iu \cdot \mathcal{C}_B} p_B \right| \leq C(1 + |u|)^{-s}$$

for $|u| \geq 1$, uniformly in the direction $u \in E^*$. On every compact annulus $\delta \leq |u| \leq M$ the left side is at most $1 - c_{\delta, M}$.

Proof. Partition the unit sphere in E^* according to a nonvanishing minor of $D\mathcal{C}_B$. On each piece use the corresponding d block coordinates as coarea coordinates. The push-forward density is $W^{s,1}$ with a uniform norm by Lemma 1.2; s integrations by parts in the push-forward variable give the first estimate. This integration includes position coordinates in position-dominated directions, so velocity integration is never asked to detect a position-only phase. Compactness and full-dimensional support imply strict modulus below one on the annulus; the uniform density bounds make the gap uniform. $\square$

1.3. A global characteristic majorant without $O(N)$ derivatives. Let $\Phi_{N,H,a}(u)$ be the centered characteristic function under the exact canonical saddle law. Successively integrate the fresh blocks selected in Lemma 1.2. At each step only derivatives in the current fixed-size block are used.

Theorem 1.4 (Blockwise global Fourier estimate). *Uniformly in the source and target charts, there are $c, C, \delta > 0$ such that*

$$\begin{aligned} |u| \leq \delta : & \quad |\Phi_N(u)| \leq C e^{-cN|u|^2}, \\ \delta < |u| \leq e^{cN} : & \quad |\Phi_N(u)| \leq C e^{-cN}, \\ |u| > e^{cN} : & \quad |\Phi_N(u)| \leq C(1 + |u|)^{-s\alpha N/2}. \end{aligned}$$

In particular, for one fixed $s_0 > d + 4$,

$$|\Phi_N(u)| \leq C e^{-cN} (1 + |u|)^{-s_0} + C(1 + |u|)^{-s_0 - cN}$$

outside a neighborhood of zero. All derivatives used in the proof have order at most the fixed number s in each block.

Proof. The small-frequency line follows from the uniform covariance lower bound and a fourth-cumulant estimate obtained by differentiating the B2 connected pressure. On a compact annulus, condition and integrate one fresh block at a time. Lemma 1.3 gives a factor at most $1 - c$; iterating through αN blocks gives e^{-cN} . For large $|u|$, the same sequential Fubini argument uses the polynomial factor $(1 + |u|)^{-s}$ from each block. The derivative is always

taken only in the current block conditional density; no mixed derivative across all blocks is generated. The exceptional event is covered by the remaining block charts rather than bounded by a frequency independent number: the measurable greedy selection partitions the entire regular configuration space into finitely many chart families, each with at least $\alpha N/2$ full-rank blocks. The singular chart boundaries have zero canonical measure and are absorbed by a smooth partition of unity. Combining the finite atlas gives the final two-line majorant. $\square$

1.4. Mixed lattice–continuous coefficient asymptotics. Separate the genuinely lattice coordinates of the constraint from the continuous quotient coordinates, writing $E = E_{\mathbb{Z}} \oplus E_{\mathbb{R}}$. For a target a in the compact interior mean image, let $\Sigma_{H,a}$ be the covariance at the source-dependent saddle. An admissible shell is

$$S_N(k, D) = \{C_N^{\mathbb{Z}} = k, N^{-1/2}(C_N^{\mathbb{R}} - Na_{\mathbb{R}}) \in D\},$$

where D has uniformly C^2 boundary and positive Gaussian mass. Shrinking families are allowed when their Gaussian mass dominates the uniform inversion error. No exact point event is asserted for a continuous coordinate.

Theorem 1.5 (Uniform local coefficient theorem). *Uniformly in H , a , admissible lattice targets, and admissible shells,*

$$\mathbb{P}_{H,\lambda(H,a)}(S_N(k, D)) = \frac{e^{-\frac{1}{2N}z_{\mathbb{Z}}^T \Sigma_{\mathbb{Z}}^{-1} z_{\mathbb{Z}}}}{(2\pi N)^{d_{\mathbb{Z}}/2} \sqrt{\det \Sigma_{\mathbb{Z}}}} [\gamma_{\Sigma_{\mathbb{R}}}(D) + o(1)].$$

Before tilting, the right side is multiplied by the exact Cramér factor $e^{-NI_H(a)}$. The theorem remains valid with a bounded regular path/contact insertion and with two source derivatives.

Proof. Use Fourier inversion on the torus for the lattice coordinates and on $E_{\mathbb{R}}^*$ for the continuous coordinates. The small ball gives the Gaussian by the cubic pressure expansion. The complement is integrable and $o(N^{-d/2})$ by Theorem 1.4. Approximate $\mathbf{1}_D$ from inside and outside by smooth functions supported in a boundary layer of width h_N . Gaussian perimeter bounds make the layer mass $O(h_N)$; choose h_N larger than the inversion error and smaller than the shell inradius. Relative error is claimed only when the Gaussian shell mass is larger than this absolute error. A path/contact insertion is a bounded source derivative of the same characteristic function, so the estimates are unchanged. $\square$

1.5. Exact-number extraction from the B2 pressure. Let $\mathcal{Q}(H, z, \lambda)$ be the B2 grand-canonical pressure with activity z . At the desired density and constraint, solve jointly

$$\partial_{\log z} \mathcal{Q} = \rho, \quad D_{\lambda} \mathcal{Q} = a.$$

Strict convexity on $\mathbb{R} \oplus E^*$ gives the unique saddle $(z(H, a), \lambda(H, a))$.

Theorem 1.6 (Source-uniform exact-number pressure). *The exact coefficient satisfies*

$[z^N]Z_\varepsilon(H, z, \lambda) = \exp\{\mu_\varepsilon \mathcal{Q}(H, z_*, \lambda_*) - N \log z_* - NI_H(a) + o(\mu_\varepsilon)\} \mathcal{G}_{\varepsilon, N}(H, a)$, where $\mathcal{G}_{\varepsilon, N}$ has the Gaussian lattice-continuous asymptotic of Theorem 1.5, locally uniformly with two source derivatives. Consequently the exact-number and microcanonical pressures exist and are obtained by the joint Legendre transform, not assumed in advance.

Proof. Use Cauchy’s formula in the activity and Fourier inversion in the constraints. Move both contours through the real saddle. The B2 normal analytic series supplies a common complex neighborhood and a uniform cubic expansion. The activity direction is handled by the same fixed-size particle-number block, and Theorem 1.4 controls the remaining Fourier directions. The local integral is the joint Gaussian determinant; every complementary contour is exponentially smaller. Uniform source derivatives follow by marking the convergent B2 clusters. Taking the logarithm and dividing by the speed gives the claimed pressures. $\square$

Corollary 1.7 (Microcanonical transfer and Schur covariance). *For every B2 path/contact source, conditioning on an admissible exact-number microcanonical shell transfers the grand-canonical Laplace principle. The conditioned covariance is*

$$\Sigma_{YY} - \Sigma_{YC} \Sigma_{CC}^{-1} \Sigma_{CY}$$

on the quotient constraint space, and all source derivatives are evaluated at the reoptimized saddle $\lambda(H, a)$.

Proof. Apply Theorem 1.6 to numerator and denominator. The same local coefficient cancels, leaving the constrained pressure. Differentiate the saddle equations with respect to the source. The implicit derivative of $\lambda(H, a)$ produces exactly the displayed Schur complement. Since Σ_{CC} is uniformly positive on E , the formula is well typed. $\square$

Theorem 1.8 (Round-twenty-one B1 closure). *The positive exact- N ensemble has a rank-correct global saddle, a fixed-size block Fourier majorant covering velocity and position constraints, a precise mixed lattice-continuous shell theorem, and a source-uniform coefficient extraction from the B2 grand-canonical pressure. No C^4 density is used for $O(N)$ integrations by parts.*

Proof. Combine Theorems 1.1, 1.4, 1.5, and 1.6. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Twenty-One dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal

reproducibility gates and do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: `qiqian@pku.edu.cn`